

Supplementary Material for “Prefix-to-Leaf: Deferred Complete-Key Selection for Semantic-ID Retrieval”

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Abstract

This document gives the model configuration, complete task results, statistical methods, and additional controls for P2L. It documents the default Independent-Suffix realization together with fixed-score and autoregressive controls that separate scoring from the selection schedule. Section S1 defines the setup; Sec. S2 gives all-task results; Sec. S3 reports candidate-budget, AR, numerical, and controlled-comparison details; and Sec. S4 gives the default timing protocol and scorer-independent index resources.

Comparison configurations. **Default** denotes Independent-Suffix P2L with the matched checkpoint, index, and realized query/candidate inputs. The AR control uses its own released checkpoint and fixes the autoregressive path score. The implementation’s optional quantizer-derived score is disabled throughout.

S1 Configuration and Statistical Methods

S1.1 Workloads and Model Settings

Tables S1 and S2 describe the workloads and masked-predictor configuration. The model dimensions and vocabulary sizes follow the released configuration and predictor definition. Dataset citations in Table S1 credit the original collections; task definitions, splits, and counts follow M-BEIR and the evaluated GENIUS pipeline [1, 2]. The frozen encoders are CLIP-based [3]; the quantizer and RQ Combiner follow GENIUS [2], using residual quantization [4]. The predictor uses the block-causal noisy/clean-context layout of block diffusion [5]; its actual training objective is specified below. Optimization uses AdamW [6].

Table S1: M-BEIR configurations. Test and LOCAL denote the numbers of test queries and task-specific candidates, respectively.

Dataset-task	Query $\rightarrow$ Candidate	Test	LOCAL
VisualNews-0 [7]	$q_t \rightarrow c_i$	19,995	542,246
MSCOCO-0 [8]	$q_t \rightarrow c_i$	24,809	5,000
Fashion200K-0 [9]	$q_t \rightarrow c_i$	1,719	201,824
WebQA-1 [10]	$q_t \rightarrow c_t$	2,455	544,457
EDIS-2 [11]	$q_t \rightarrow (c_i, c_t)$	3,241	1,047,067
WebQA-2	$q_t \rightarrow (c_i, c_t)$	2,511	403,196
VisualNews-3	$q_i \rightarrow c_t$	20,000	537,568
MSCOCO-3	$q_i \rightarrow c_t$	5,000	24,809
Fashion200K-3	$q_i \rightarrow c_t$	4,889	61,707
NIGHTS-4 [12]	$q_i \rightarrow c_i$	2,120	40,038
OVEN-6 [13]	$(q_i, q_t) \rightarrow c_t$	50,004	676,667
InfoSeek-6 [14]	$(q_i, q_t) \rightarrow c_t$	11,323	611,651
FashionIQ-7 [15]	$(q_i, q_t) \rightarrow c_i$	6,003	74,381
CIRR-7 [16]	$(q_i, q_t) \rightarrow c_i$	4,170	21,551
OVEN-8	$(q_i, q_t) \rightarrow (c_i, c_t)$	14,741	335,135
InfoSeek-8	$(q_i, q_t) \rightarrow (c_i, c_t)$	17,593	481,782

Training uses the released upsampled union of 1,332,790 entries per epoch, shuffled at the entry level without an additional task-specific sampler. Weight decay excludes biases, normalization/gain parameters, logit scale, and other one-dimensional parameters. Training uses one GPU; training over 100 epochs totals 9.46 GPU-hours. The augmentation and loss are specified below.

S1.2 Training Objective

The evaluated checkpoint uses the training-only query-positive augmentation of GENIUS [2]. Let $\mathbf{z}_q$ and $\mathbf{z}_d$ be the normalized retrieval embeddings of the query and its positive candidate. For every feature dimension, $u_k \sim \text{Beta}(2, 2)$,

Table S2: Masked-predictor architecture, training, and default inference settings. The AR control uses a separate released GENIUS checkpoint.

Architecture / training	Value	Training / inference	Value
Retrieval encoder	CLIP ViT-L/14, frozen	Optimizer / learning rate	AdamW / 10^{-4}
RQ Combiner / quantizer	Frozen / RQ50	Schedule / weight decay	Cosine / 10^{-4}
Semantic-ID length L	9	Training duration	100 epochs (0-99)
Predictor layers / heads	6 / 8	Training / inference batch size	512 / 8
Hidden / feed-forward size	512 / 2,048	Training partition	Nine singleton blocks
Query-memory shape	30×512	P2L suffix visibility	Independent-Suffix; one shared guided round
Predictor dropout	0.1	Prefix beam / final-key budget (B, K)	(50, 50)
Modality vocabulary	3	Switch depth s	3, including y_0
Residual vocabularies	8 levels of 4,096	CFG scale ω	3.0
Classifier	Tied; MASK input-only	Masked checkpoint	Epoch 99
Mask timesteps T	8	Inference timestep	$t_{\text{inf}} = 8$ each round
Trainable parameters	53,857,280	Visible-position weight η	0.1
Conditional-dropout rate	0.1	Random seed	2023

and the projector condition is $\mathbf{r}_c = \text{norm}(\sqrt{\mathbf{u}} \odot \mathbf{z}_q + \sqrt{1 - \mathbf{u}} \odot \mathbf{z}_d)$. This element-wise mixing is applied to every training example; the prediction target remains the candidate identifier $y(d)$, and inference uses the query embedding alone.

The query projector maps the normalized conditioning vector to cross-attention memory:

$$\mathbf{H}_q = \text{reshape} \left(\mathbf{W}_q \frac{\mathbf{r}_c}{\max(\|\mathbf{r}_c\|_2, \epsilon)} + \mathbf{b}_q, N_q, d \right). \quad (\text{S1})$$

Here $N_q = 30$, $d = 512$, $\mathbf{W}_q \in \mathbb{R}^{N_q d \times d_r}$, and $\mathbf{b}_q \in \mathbb{R}^{N_q d}$. The memory supplies cross-attention keys and values; inference sets $\mathbf{r}_c = \mathbf{r}_q$.

For each example n in minibatch $\mathcal{A}$, the timestep t_n is sampled uniformly from $\{1, \dots, T\}$. Each valid position is independently masked with probability t_n/T . The selected positions form $\mathcal{M}_n$; if it is empty, one uniformly sampled valid position is added. The modality code follows the same process. For the resulting input $\tilde{\mathbf{y}}^{(n, t_n)}$, let $w_{nj} = 1$ on masked positions and $w_{nj} = \eta$ on visible valid positions. Then the implementation minimizes the batch-globally normalized objective

$$\mathcal{L}_{\text{mask}} = - \frac{\sum_{n \in \mathcal{A}} \sum_{j \in \mathcal{I}_{\text{valid}}^{(n)}} w_{nj} \log p_{\theta}(y_j^{(n)} | \tilde{\mathbf{y}}^{(n, t_n)}, \mathbf{r}_c^{(n)}, t_n)}{\sum_{n \in \mathcal{A}} \sum_{j \in \mathcal{I}_{\text{valid}}^{(n)}} w_{nj}}. \quad (\text{S2})$$

Masked positions receive full reconstruction weight, while visible positions retain the smaller weight η used by the evaluated checkpoint.

The checkpoint is trained with singleton-style attention visibility and complete-leaf ranking is not part of optimization. Sequential inference commits one additional code per round, so each later prediction consumes the newly committed suffix. Default P2L instead evaluates every unresolved suffix position from the same retained prefix in one guided round and masks cross-suffix-state visibility. Relative to Sequential, the matched study therefore changes both the execution/selection schedule and suffix conditioning; it does not introduce an interacting suffix block.

For classifier-free guidance [17], conditional dropout replaces the query-conditioning memory with a learned null condition with probability 0.1. At inference the two logit streams use the same timestep and are combined as $\ell^{\text{cfg}} = \ell^u + \omega(\ell^c - \ell^u)$, with $\omega = 3$. Training completed 260,182 optimizer updates over 100 epochs.

S1.3 Attention Visibility in the Default Scorer

The main paper gives the exact inference visibility mask used by Independent-Suffix scoring. The training-time singleton partition uses the same noisy/clean stream structure but does not impose the additional committed-prefix restriction on clean keys. At inference, uncommitted clean placeholders are hidden, so unresolved positions cannot communicate directly or indirectly through clean states. This architectural restriction supports reusable prefix-conditioned suffix distributions; it is not a statistical independence assumption about the data.

Regression checks cover attention visibility, cross-suffix perturbations, agreement between shared and repeated singleton guidance, and legal-key top- K selection.

S1.4 Metrics and Statistical Conventions

Let R_q be the set of relevant items for query q and $T_q^{(k)}$ the first k retrieved items. Following the evaluated benchmark implementation, item-level $R@k$ is a query hit rate:

$$R@k = \frac{100}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}[R_q \cap T_q^{(k)} \neq \emptyset]. \quad (\text{S3})$$

With multiple positives, retrieving any one counts as a hit; this metric is not the fraction of all relevant items retrieved. $IDR@k$ instead tests whether one of the first k returned semantic IDs has a relevant item in its exact-ID bucket. Relevant-frontier survival tests whether $\mathcal{F}_s$ contains at least one relevant item’s ID before final key selection. The reported leaf count refers to distinct complete keys, not expanded items.

The 16-task matched macro assigns equal weight to each task’s $R@10$, including the fashion tasks. The separate UNION control also reports $R@10$ for both evaluated tasks, using the benchmark relevance labels [1, 2]. All gains are percentage-point differences unless stated otherwise. Where unrounded values are available, differences are computed before display rounding, so displayed columns can differ from their displayed difference by 0.01 points. Default matched scores are calculated from integer hit counts; display values use decimal half-up rounding only at the final step.

The evaluation-only twelve are the primary effectiveness group not used for policy selection. The four development configurations support diagnostic comparisons; their union with the evaluation-only twelve is a descriptive all-16 macro. The per-query candidate analysis pairs outputs by query ID and counts P2L-only and Sequential-only hits at $R@10$. For each task, their difference divided by the query count is the $R@10$ gain; both-hit and neither-hit queries contribute zero.

All intervals use the paired-query bootstrap [18]. For the default Independent-Suffix comparison, the archived joint counts are ordered as (neither hit, P2L only, Sequential only, both hit). We draw 10,000 multinomial resamples per configuration using one NumPy `default_rng(2023)` stream in the row order of Table S4. This is distributionally equivalent to resampling query pairs for binary hit metrics. The 2.5th and 97.5th percentiles use linear interpolation. A group’s interval is calculated from the unweighted mean of its task draws at each replicate. Table S4, the main paper’s Figure 3, and the grouped summaries share this single recomputation. The script and source hashes are recorded in `evidence/derived/bootstrap_protocol.json`.

The Eval-12 result is +3.29 points [3.07, 3.52]; Dev-4 gives +7.24 [6.50, 7.97]; All-16 gives +4.28 [4.03, 4.53]. The grouping follows the documented policy-development chronology rather than outcome-based task selection; the grouped macro itself was not preregistered. The separately archived AR intervals use 10,000 paired resamples with seed 2023; the Eval-12 fixed-score control uses seed 20260914. Those intervals remain as reported and are not replaced by the default-comparison bootstrap. All intervals condition on the evaluated checkpoints, queries, and task configurations; they do not measure future-task or training-seed variability, and task-level intervals have no multiple-comparison adjustment.

Query-level leaf percentiles and latency percentiles have different sampling units. For batch-amortized timing, each synchronized batch duration is divided by its batch size; the reported percentiles do not represent independent batch-1 requests. The separately profiled batch-1 timings are identified in the default timing tables below. Reported generation time includes neural evaluation, Trie operations, score aggregation, and final ID selection, but excludes offline preparation, index loading, raw-query encoding, bucket expansion, and item reranking.

S2 Matched LOCAL Decoder Results

The evaluation-only twelve are the primary effectiveness group excluded from P2L policy selection. Within this matched comparison, the query inputs, checkpoint, semantic-ID index, and reranker are fixed while Sequential is replaced by masked P2L. Independent-Suffix P2L was fixed using only CIRRR-7, NIGHTS-4, EDIS-2, and WebQA-1 before evaluation was extended to the remaining twelve configurations. On this evaluation-only group, macro $R@10$ is 38.81 for Sequential and 42.10 for P2L, a +3.29-point difference, with all twelve task-level point estimates positive.

For completeness, Table S4 reports the same matched comparison over all sixteen configurations, all evaluated at $R@10$ here, including the fashion tasks. The descriptive all-16 macro is 39.36 versus 43.64 (+4.28 points); all sixteen point estimates are positive and fourteen paired bootstrap intervals exclude zero. Its macro 95% interval is [4.03, 4.53]. Interacting-Suffix P2L is retained as a later exploratory implementation ablation with macro $R@10$ 42.83 and did not change the default policy.

Table S3: Evaluation-group R@10 summaries (%). Eval-12 is the primary group excluded from policy selection; Dev-4 and All-16 are descriptive diagnostics. Intervals resample paired queries within fixed configurations.

Group	Seq.	P2L	Δ	95% CI
Dev-4	40.99	48.24	+7.24	[6.50, 7.97]
Eval-12	38.81	42.10	+3.29	[3.07, 3.52]
All-16	39.36	43.64	+4.28	[4.03, 4.53]

Table S4: Matched LOCAL results on all 16 M-BEIR configurations. P2L uses the default Independent-Suffix implementation. Brackets are 95% paired-query bootstrap intervals for the R@10 difference, recomputed from the archived joint hit counts.

Task	Seq. R@10	P2L R@10	Δ R@10	95% CI
VisualNews-0	22.36	29.22	+6.87	[6.36, 7.38]
MSCOCO-0	82.65	83.41	+0.75	[0.60, 0.91]
Fashion200K-0	13.79	14.72	+0.93	[-0.47, 2.27]
WebQA-1	32.26	46.60	+14.34	[12.30, 16.33]
EDIS-2	40.98	50.73	+9.75	[8.15, 11.35]
WebQA-2	56.15	65.03	+8.88	[7.21, 10.55]
VisualNews-3	23.30	30.68	+7.38	[6.85, 7.90]
MSCOCO-3	93.40	94.22	+0.82	[0.36, 1.30]
Fashion200K-3	15.52	16.12	+0.59	[-0.16, 1.35]
NIGHTS-4	48.58	50.28	+1.70	[0.66, 2.74]
OVEN-6	40.73	43.50	+2.76	[2.54, 2.98]
InfoSeek-6	21.70	23.81	+2.11	[1.65, 2.58]
FashionIQ-7	15.16	17.39	+2.23	[1.48, 3.00]
CIRR-7	42.16	45.35	+3.19	[2.23, 4.17]
OVEN-8	45.95	50.44	+4.49	[4.06, 4.92]
InfoSeek-8	35.03	36.69	+1.66	[1.28, 2.05]
Task macro	39.36	43.64	+4.28	[4.03, 4.53]

S2.1 Reproduced Reference Baselines

The main paper reports fresh Eval-12 LOCAL reproductions of CLIP-SF, GENIUS, and GENIUS^R using a uniform item-level R@10. All three use the same official LOCAL candidate pools and the same twelve evaluation-only configurations, but they use their own released checkpoints and therefore serve as reference systems rather than matched execution controls. CLIP-SF performs exact inner-product search over frozen CLIP-SF embeddings. GENIUS and GENIUS^R reuse the same generated identifier lists (beam width 50, FP16, inference batch size 8); GENIUS performs exact-ID bucket lookup without embedding reranking, while GENIUS^R adds the original embedding reranking stage. Their Eval-12 macro R@10 scores are 54.63, 31.88, and 38.49, respectively. Normalized reproduction summaries and checkpoint fingerprints are included under `evidence/baseline_reproduction_20260916/`. The matched Sequential and P2L macros remain 38.81 and 42.10; these share the checkpoint, semantic-ID index, query/candidate realization, and reranker.

Table S5: Exploratory suffix-visibility comparison on all 16 LOCAL test configurations. Both P2L variants use the same masked checkpoint, $s = 3$ and $B = K = 50$. R@10 is in percent; Δ is Independent-Suffix minus Interacting-Suffix in percentage points. The macro difference is computed before rounding.

Task	Interacting-Suffix	Independent-Suffix	Δ
VisualNews-0	29.59	29.22	-0.37
MSCOCO-0	83.36	83.41	+0.05
Fashion200K-0	14.95	14.72	-0.23
WebQA-1	43.83	46.60	+2.77
EDIS-2	50.17	50.73	+0.56
WebQA-2	61.69	65.03	+3.34
VisualNews-3	30.78	30.68	-0.10
MSCOCO-3	94.08	94.22	+0.14
Fashion200K-3	16.22	16.12	-0.10
NIGHTS-4	50.28	50.28	0.00
OVEN-6	42.50	43.50	+1.00
InfoSeek-6	22.27	23.81	+1.54
FashionIQ-7	16.26	17.39	+1.13
CIRR-7	44.34	45.35	+1.01
OVEN-8	49.56	50.44	+0.88
InfoSeek-8	35.45	36.69	+1.24
Macro	42.83	43.64	+0.80

S3 Mechanism and Candidate-Budget Controls

The equal-item-budget control uses saved outputs from the default Independent-Suffix comparison. A fixed-score control uses the default masked checkpoint while changing only the selection schedule; the autoregressive study repeats the selection test with its own released AR scorer.

S3.1 Equal Item Budget

Table S6: Default Independent-Suffix P2L and Sequential, $B = K = 50$. Each policy retains the first 50 distinct expanded items before reranking. R@10 values are percentages; differences use unrounded values.

Task	Policy	Mean items	Full R@10	Cap-50 R@10	Cap minus full
CIRR-7	Sequential	50.97	42.16	42.13	-0.02
	P2L	51.00	45.35	44.89	-0.46
NIGHTS-4	Sequential	50.08	48.58	48.58	0.00
	P2L	50.08	50.28	50.28	0.00
EDIS-2	Sequential	51.42	40.98	40.91	-0.06
	P2L	51.58	50.73	50.29	-0.43
WebQA-1	Sequential	50.22	32.26	32.26	0.00
	P2L	50.24	46.60	46.48	-0.12

This analysis uses saved query beams, embeddings, candidate codes, item IDs, and official qrels. The paired inputs are identical within each task; full-expansion R@10 reproduces the main comparison to its reported precision. We expand buckets in beam order, preserve candidate-array order within a bucket, deduplicate items, and keep the first 50 items before cosine reranking. Scores are recomputed in FP32 with stable tie handling. P2L’s largest cap-induced decrease is 0.46 points on CIRR-7. Its positive gain persists on all four tasks, although bucket expansion contributes to the magnitude of some gains.

S3.2 Default-Checkpoint Fixed-Score Selection Control

Frozen-Levelwise and Frozen-P2L use the same Independent-Suffix position-wise log-probabilities and sum aggregation after the common depth-three prefix stage. They also share the checkpoint, query inputs, candidate codes, qrels, quantizer, reranker, deterministic tie rule, and $B = K = 50$. Frozen-Levelwise applies legal top- B pruning after each unresolved position; Frozen-P2L scores stored complete descendants before the next selection. Production-path regression checks verify equal prefix frontiers, guidance inputs, and suffix logits on deterministic fixtures. Thus this comparison changes the intermediate selection schedule while holding the complete-key scoring rule and position-wise score tables fixed.

Development diagnostic. The original four-workload diagnostic was run only on the configurations used to choose the fixed execution policy. It establishes the mechanism during development; the twelve evaluation-only configurations provide the separate selection-control extension.

Table S7: Development fixed-score selection diagnostic (%). Frozen-LW is legal level-wise top- B selection and Frozen-P2L is deferred complete-descendant selection. R@10 intervals use 10,000 paired-query bootstrap resamples. Macro values are unweighted task means.

Task	R@10				IDR@50			Queries
	Frozen-LW	Frozen-P2L	Δ	95% CI	Frozen-LW	Frozen-P2L	Δ	
CIRR-7	42.21	45.35	+3.14	[2.16, 4.10]	51.22	54.05	+2.83	4,170
NIGHTS-4	49.10	50.28	+1.18	[0.14, 2.22]	76.98	81.70	+4.72	2,120
EDIS-2	39.15	50.73	+11.57	[10.00, 13.14]	43.29	56.40	+13.11	3,241
WebQA-1	29.90	46.60	+16.70	[14.75, 18.58]	30.47	47.13	+16.66	2,455
Task macro	40.09	48.24	+8.15	[7.42, 8.87]	50.49	59.82	+9.33	–

Evaluation-only selection-control campaign. After fixing the scoring and execution configuration, we repeated the selection-only comparison on all twelve configurations excluded from policy development. The campaign contains 178,587 paired test queries. Table S8 reports the full task-level results. “Survival” is the fraction of queries for which at least one relevant complete key lies in the depth-three exposed frontier. APF (avoidable pruning failure) is the fraction of queries for which a relevant complete key lies in that frontier and is selected by Frozen-P2L but not by Frozen-Levelwise.

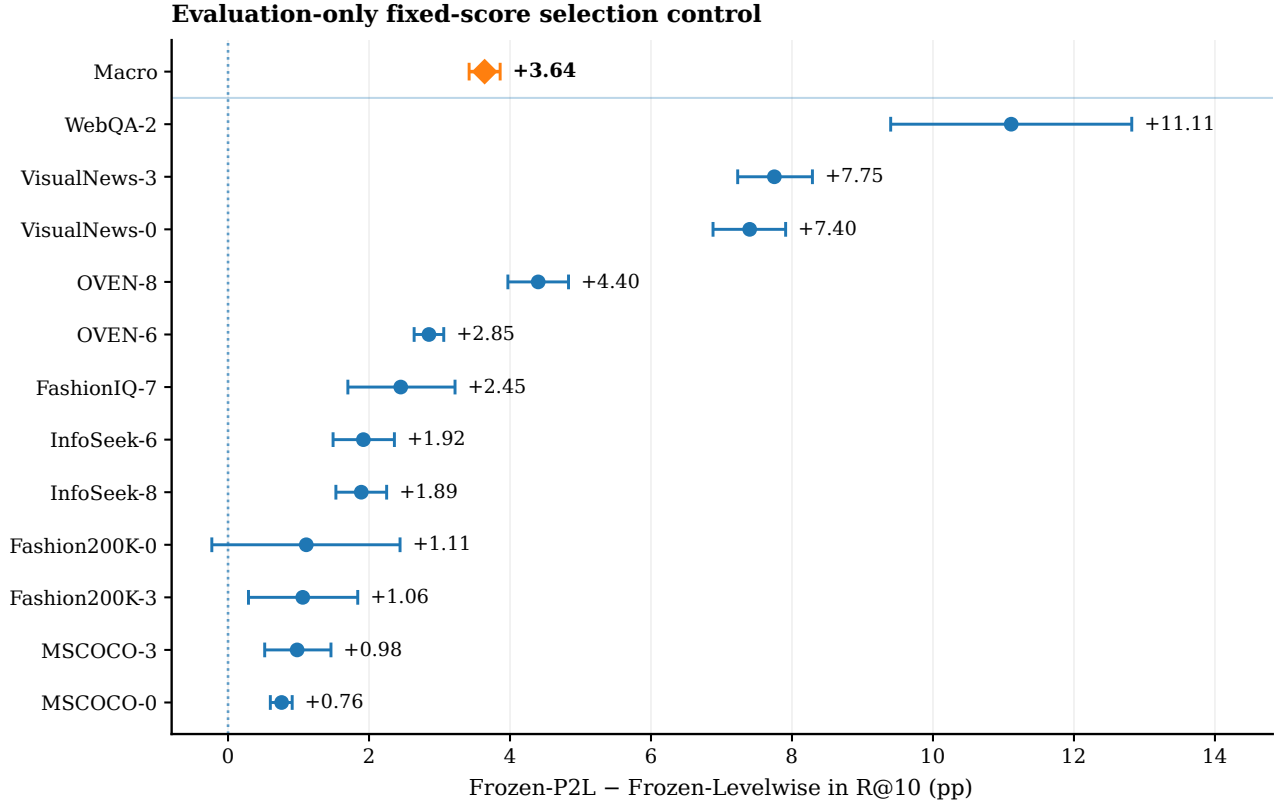


Figure S1: Evaluation-only fixed-score selection effect with 95% paired-query bootstrap intervals. Every point estimate is positive; only Fashion200K-0 crosses zero. The macro interval is shown separately at the top.

Frozen-P2L improves macro R@10 by 3.64 points with paired-query 95% CI [3.42, 3.86], and all 12/12 task point estimates are positive. Eleven task intervals exclude zero; Fashion200K-0 remains a positive but non-significant estimate. Macro IDR@50 improves by 4.46 points. The 7.63% task-macro APF shows that avoidable post-switch pruning failures are non-negligible across the evaluation-only configurations. APF is a failure-mode diagnostic, not an additive decomposition of the R@10 gain.

The campaign also provides an execution cross-check under the same score configuration. Across all 178,587 queries, total generation time is 4,245.2 s for Frozen-P2L and 10,147.8 s for Frozen-Levelwise. Dividing by the total query count gives query-weighted averages of 23.77 and 56.82 ms/query and a 58.2% reduction. Downstream retrieval takes 2,287.0

Table S8: Evaluation-only fixed-score selection control. R@10, IDR@50, Survival, and APF are percentages. All twelve R@10 point estimates favor Frozen-P2L; only Fashion200K-0 has a task-level 95% interval crossing zero.

Task	Queries	Frozen-LW R@10	Frozen-P2L R@10	Δ	95% CI	Frozen-LW IDR@50	Frozen-P2L IDR@50	Survival	APF
VisualNews-0	19,995	21.82	29.22	+7.40	[6.88, 7.91]	23.91	32.11	56.91	12.18
MSCOCO-0	24,809	82.65	83.41	+0.76	[0.60, 0.91]	88.79	90.01	91.06	1.66
Fashion200K-0	1,719	13.61	14.72	+1.11	[-0.23, 2.44]	21.06	24.49	75.10	9.31
WebQA-2	2,511	53.92	65.03	+11.11	[9.40, 12.82]	56.39	67.70	88.89	16.21
VisualNews-3	20,000	22.92	30.68	+7.75	[7.23, 8.29]	25.00	33.92	57.79	12.67
MSCOCO-3	5,000	93.24	94.22	+0.98	[0.52, 1.46]	96.80	97.74	99.14	1.56
Fashion200K-3	4,889	15.05	16.12	+1.06	[0.29, 1.84]	22.72	25.79	68.01	8.35
OVEN-6	50,004	40.65	43.50	+2.85	[2.64, 3.06]	43.75	47.39	57.83	5.38
InfoSeek-6	11,323	21.89	23.81	+1.92	[1.49, 2.36]	26.56	28.92	45.55	4.91
FashionIQ-7	6,003	14.94	17.39	+2.45	[1.70, 3.22]	20.51	23.64	57.85	8.28
OVEN-8	14,741	46.04	50.44	+4.40	[3.97, 4.83]	47.36	52.09	60.86	6.22
InfoSeek-8	17,593	34.80	36.69	+1.89	[1.53, 2.25]	39.14	41.65	57.11	4.80
Task macro	–	38.46	42.10	+3.64	[3.42, 3.86]	42.67	47.12	68.01	7.63

versus 2,275.1 s, respectively, so Frozen-P2L is 0.52% slower in that stage. The speed evidence from this control therefore localizes to identifier generation rather than an unconditional end-to-end serving speedup. The exposed frontier has an unweighted task-macro mean of 1,470 complete keys, macro mean P95 of 3,321, and a largest observed task maximum of 78,585 keys. This frontier distribution motivated the tie-fixed exact-streaming audit reported in Sec. S4.1.1.

For frontier accounting only, Frozen-Levelwise query-profile files reuse the Frozen-P2L profiles because the first three prefix-localization rounds are identical at $s = 3$, $B = 50$. Frozen-Levelwise codes, scores, metrics, and timings were generated independently. First-pruned-depth trajectories were not captured in this campaign. The bootstrap uses 10,000 paired-query resamples with seed 20260914 on one RTX 3090 (24 GB). The archived experiment report and machine-readable CSV/JSON summaries are included in the submission evidence package.

The fixed-score comparison is interpreted separately from the default Sequential-P2L gap: Frozen-Levelwise isolates selection under frozen score tables, whereas the default Sequential comparison also changes suffix conditioning.

S3.3 Score-Preserving Autoregressive Control

The masked instantiation above changes the suffix conditioning context, so we also use an autoregressive (AR) instantiation as a mechanism control. It keeps the original autoregressive path score fixed and changes only when candidate pruning occurs. For a complete key, let

$$S_{\text{AR}}(y; q) = \sum_{j=0}^{L-1} \log p_{\theta}(y_j \mid y_{<j}, q) \quad (\text{S4})$$

be the sum of its token log-probabilities, without length normalization or renormalization over Trie-valid keys. AR-P2L performs ordinary Trie-constrained traversal through depth s and retains the corresponding AR prefix score, $S_{\text{pre}}^{\text{AR}}(y_{<s}; q) = \sum_{j=0}^{s-1} \log p_{\theta}(y_j \mid y_{<j}, q)$. It then supplies each exposed leaf as a shifted-right decoder input and evaluates

$$S_{\text{AR-P2L}}(y; q) = S_{\text{pre}}^{\text{AR}}(y_{<s}; q) + \sum_{j=s}^{L-1} \log p_{\theta}(y_j \mid y_{<j}, q). \quad (\text{S5})$$

Causal masking keeps the state predicting y_j from reading y_j or later codes. Teacher forcing therefore evaluates the same conditional terms as stepwise scoring: Eqs. (S5) and (S4) are equal in exact arithmetic, with numerical differences possible from floating-point roundoff. Setting $S = S_{\text{AR}}$ in the main paper’s complete-key selection rule changes selection, not the specified path score. Candidate chunks bound teacher-forcing memory. Each candidate still requires its own conditional path evaluation, so neural work grows with the exposed frontier rather than only the number of retained prefixes.

S3.4 Autoregressive Results

The AR scorer is the released GENIUS head [2], based on T5-small [19]. AR-Sequential and AR-P2L share the checkpoint, realized query tensors, native candidate codes and identifiers, candidate embeddings, exact-ID buckets, and reranker inputs within each task. Both use $B = K = 50$ and fixed nine-token keys. AR-Sequential performs deterministic Trie-constrained beam search through depth nine. AR-P2L retains the same prefix-search interface through $s = 3$, enumerates stored descendants, and teacher-forces complete leaves in memory-bounded chunks. These runs were not collected under the masked timing protocol, so the control supports retrieval quality and score preservation, not an AR efficiency claim.

Table S9: Detailed paired results for the score-preserving AR control (%). Each confidence interval uses 10,000 paired-query bootstrap resamples for AR-P2L minus AR-Sequential.

Task	Metric	AR-Sequential	AR-P2L	Δ	95% CI
NIGHTS-4	IDR@1	0.85	0.80	-0.05	[-0.24, 0.09]
	IDR@10	24.10	24.06	-0.05	[-0.57, 0.47]
	IDR@50	76.46	81.70	+5.24	[3.96, 6.51]
	R@1	6.98	7.45	+0.47	[0.05, 0.90]
	R@5	29.10	30.47	+1.37	[0.61, 2.17]
	R@10	48.73	50.47	+1.75	[0.75, 2.78]
WebQA-1	IDR@1	9.61	10.10	+0.49	[-0.33, 1.30]
	IDR@10	25.38	34.09	+8.72	[7.17, 10.26]
	IDR@50	31.81	54.09	+22.28	[20.41, 24.15]
	R@1	23.30	38.86	+15.56	[13.89, 17.27]
	R@5	30.22	51.16	+20.94	[19.14, 22.73]
	R@10	31.28	52.91	+21.63	[19.80, 23.46]

Table S10: Numerical checks for the AR backend, not additional retrieval benchmarks. FP16 near-ties can change boundary membership even when the specified autoregressive path score is unchanged.

Check and sample	Observed agreement
Stepwise versus teacher forcing: two CIRR queries, 128 leaves per query, depths 0, 3, and 6	Maximum absolute score error 3.05×10^{-5} ; all six comparisons return the same top-50 set.
Comparison with Hugging Face constrained generation: eight sampled FP16 query comparisons	Seven top-50 sets are identical; one NIGHTS query has 48/50 overlap at the beam boundary.

The first check evaluates two implementations of the same complete-path score. The second also includes finite-precision prefix and beam-boundary decisions; it is therefore not a bitwise-equivalence guarantee. A near-tie can change which prefix survives even when the specified autoregressive path score is unchanged.

S3.5 Validation Depth

The archived logs also measure the physical work at each depth (Table S11). EDIS-2 at $s = 2$ reaches 985,794 complete keys on one validation query, despite $B = K = 50$. Its mean exposed frontier drops from 871,423 keys at $s = 2$ to 10,504 at $s = 3$. This explains why saving one dependent round need not save time. Depth changes prefix context and exposure together, so these observations characterize execution sensitivity.

Table S11: Work and cost across validation depths, from hash-checked logs of the same 16 runs summarized by the main-paper validation plot. $B = K = 50$, batch size 8. Counts are complete keys; times are milliseconds per query. CPU enumeration is included in generation, not additive to it. Each point has one timing run.

Task	s	Mean keys	P99 keys	Max keys	CPU enum.	Generation	R@10 (%)
CIRR-7	2	8,824.3	13,550.2	14,962	0.57	19.54	60.40
CIRR-7	3	316.3	655.1	800	0.41	24.32	62.40
CIRR-7	4	79.5	143.0	173	0.46	29.19	60.05
CIRR-7	5	54.5	81.0	91	0.48	34.68	58.65
NIGHTS-4	2	23,740.6	32,751.9	35,097	0.44	19.94	48.83
NIGHTS-4	3	487.4	975.0	1,329	0.45	23.72	49.54
NIGHTS-4	4	101.5	206.0	255	0.52	28.88	48.67
NIGHTS-4	5	60.8	102.0	167	0.55	34.03	48.57
EDIS-2	2	871,422.5	978,586.5	985,794	40.20	156.37	45.90
EDIS-2	3	10,503.8	36,524.6	49,463	1.43	25.37	51.05
EDIS-2	4	512.9	2,469.6	3,575	1.26	29.02	46.73
EDIS-2	5	96.8	427.0	1,205	0.90	34.16	43.77
WebQA-1	2	374,312.7	458,900.9	476,915	25.28	99.61	32.24
WebQA-1	3	9,277.7	22,503.7	27,309	1.41	25.84	44.84
WebQA-1	4	452.3	1,511.5	2,055	1.20	29.88	41.40
WebQA-1	5	83.6	692.4	845	0.86	34.89	34.17

S3.6 Candidate-Width Operating Points

The main paper reports the default $B = K = 50$ operating point. Figure S2 shows all 32 measured width–quality–latency points on the four development workloads; points are measured runs and connecting lines are visual guides only.

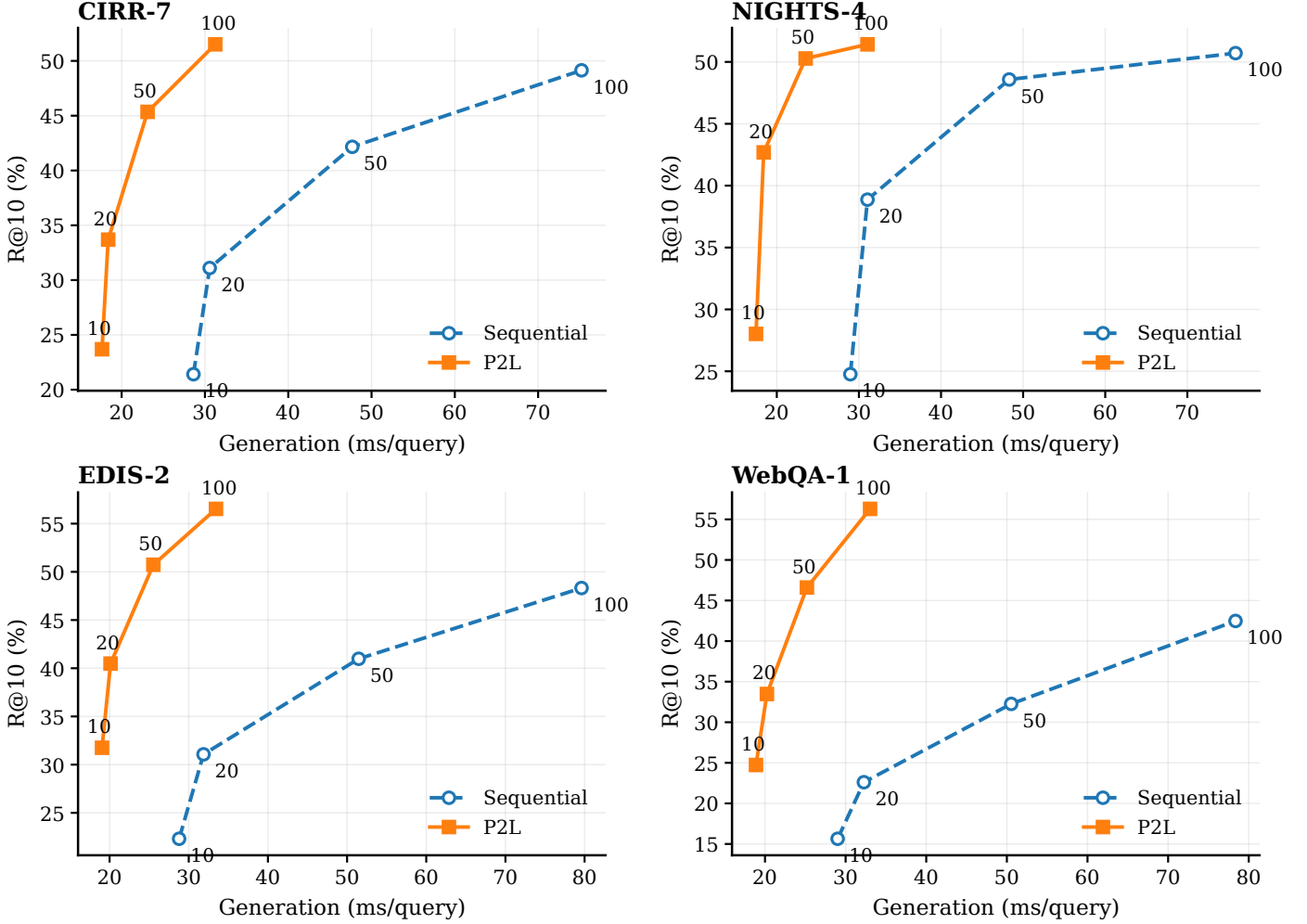


Figure S2: R@10 versus generation latency across $B = K \in \{10, 20, 50, 100\}$. P2L improves retrieval at every tested width and shifts the measured operating frontier toward higher quality and lower generation cost. Numeric annotations identify the tested $B = K$ budget.

S3.7 Comparison Contracts

Table S12: Factors held fixed and changed in each reported comparison, together with the corresponding interpretation.

Comparison	Held fixed	Changed	Interpretation
Matched LOCAL policies	Checkpoint, query realization, index, $B = K = 50$, reranker	Suffix conditioning and selection schedule	Measures the joint effect of suffix conditioning and selection scheduling under matched trained assets.
Suffix visibility	Checkpoint, task inputs, $s = 3$, $B = K = 50$	Attention links between unresolved suffix states	Compares suffix-state visibility under the fixed P2L policy; Independent-Suffix was selected on the four development workloads, and the all-16 comparison is an exploratory ablation.
Equal-item budget	Saved generated beams and embeddings; common 50-item cap	Full expansion versus common 50-item cap	Tests whether the retrieval gain persists under the same maximum reranking budget; generation is unchanged.
Fixed-score selection	Default masked checkpoint, Independent-Suffix scores, depth-three prefix setup, index, inputs, reranker	Intermediate legal top- B versus deferred complete-descendant selection	Isolates intermediate selection scheduling under fixed position-wise scoring evidence, with four development diagnostics and all twelve evaluation-only configurations.
Score-preserving AR	AR checkpoint, autoregressive path score, realized inputs, Trie, reranker	Continued pruning versus descendant selection	Tests deferred selection under a fixed autoregressive path score on two development workloads.
Validation depth	Masked checkpoint, validation tasks, $B = K = 50$	$s \in \{2, 3, 4, 5\}$	Characterizes the tested depth-quality-cost trade-off; candidate exposure and suffix context vary with s .
Repeated batch-1	Within-task inputs, hardware, single-query batches	Sequential versus default P2L	Measures generation savings with single-query batches; reported statistics exclude the first 100 requests of each run.
Matched UNION	5.61M-item index, within-task inputs, $B = K = 50$	Sequential versus default P2L on two tasks	Tests the frozen policy on a 5.61M-item shared collection for two workloads.

S3.8 Execution-Mechanism Comparison with WIDE

Table S13 separates the algorithms at the level of query execution. The WIDE entries summarize its published method; no matched WIDE quality or latency measurements are reported here.

Table S13: Conceptual execution comparison. WIDE is summarized from its published algorithm [20]; no WIDE number is combined with the matched P2L experiments. “Complete competition” asks which stored candidates can compete at the plan’s final ranking stage.

Plan	Intervention	Trigger	Expanded set	Candidate score	Complete competition
Sequential	Trie top- B pruning	Every identifier level	Legal children of retained prefixes	Cumulative level-wise generation score	Limited by every earlier top- B decision
WIDE	Wildcard at uncertain levels; later AR generation can resume	Query/beam entropy above an offline-calibrated level threshold	Candidates matching confident, non-wildcard positions	Hybrid of reliable generation confidence and continuous similarity	Over the wildcard-expanded candidate pool
P2L	Stop intermediate selection after prefix localization	Fixed common depth s	Every stored descendant of the retained depth- s prefixes	Complete-key score; the default masked realization uses a prefix score plus shared suffix-position scores	Global top- K over the complete-key frontier

S4 Default Timing Protocol and Index Resources

S4.1 Timing Protocol

The reported default system evidence uses Independent-Suffix P2L. The 5.61M-item UNION control uses matched policy runs; the batch-1 study uses five repeats. Generation begins from precomputed query embeddings and includes query projection, neural prediction, Trie operations, suffix aggregation, and identifier selection. D2R adds bucket expansion and embedding reranking. Both exclude raw-query encoding and offline index preparation/loading.

Both studies use an NVIDIA GeForce RTX 3090, driver 550.144.03, Python 3.10.19, PyTorch 2.7.0, and CUDA 12.8 on Linux. In the standardized batch-1 campaign, CUDA events record generation and its GPU stages, while CPU wall-clock timers record per-query bucket lookup and reranking. D2R is their additive pipeline interval and is not a single raw-input request timer. All reported per-query latency and stage statistics exclude the first 100 requests of each run; accuracy, frontier-count, and run-level memory statistics use the complete evaluated run. Mean $\pm$ sample standard deviation is computed over five per-run means; P95 is the mean of five per-run percentiles, not a pooled percentile. Policy order alternates by repeat and task order rotates. Retrieval accuracy is identical across repeats within each policy.

The outer generation events are recorded immediately before and after the model call, before output collection in the evaluation loop. CPU continuation lookup and assembly, host-to-device transfers, and GPU scoring occur between these events. Elapsed stream time includes host-induced device idle gaps but is not an independent CPU wall-clock measurement. The data-loader wait occurs before the start event; final loop-level output collection occurs after the end event. GPU synchronization precedes the generation loop and follows its completion before event durations are read. The batch-amortized generation mean instead uses the synchronized wall-clock duration of the full generation loop divided by query count; these two timing granularities are reported separately.

The continuation cache maps a (prefix, suffix-length) pair to a host tensor of legal codes. A miss traverses the nested Trie; a hit reuses the tensor. Parent-row arrays, transfers, query-conditioned scores, and final selection are still evaluated for each request. Each repeat launches a fresh evaluation process, so the cache persists across queries, not across repeats. Excluding the first 100 queries does not guarantee that every later prefix is cached: later misses remain in the measured stream. The separately reported enumeration timer covers continuation lookup and parent-row construction, not the subsequent concatenation and transfer. For D2R, generation and retrieval durations are paired by query and added before computing each run’s P95; summing stage P95 values would be a different statistic.

Batch-1 and batch-8 measurements share query IDs and candidate codes but use separately realized query embeddings. Across the four tasks, their maximum absolute embedding difference is 9.16×10^{-4} . Accuracy repeatability is assessed within each batch setting; comparisons across settings test whether generation savings persist.

S4.1.1 Tie-Fixed Exact Streaming Top- K Audit

The default timing results above use Full materialization of the exposed complete-key frontier. We additionally audit a bounded-buffer Streaming realization on the twelve evaluation-only tasks. Both Full and Streaming use the same frozen checkpoint, semantic-ID index, queries, runtime environment, and canonical lexicographic ordering: complete-key score descending, followed by semantic-ID/leaf order ascending. Streaming scores descendants in chunks and merges each chunk into a running top- K under this same total order.

The campaign contains 48 completed cells: 12 Full-reference runs and 36 Streaming runs using chunk sizes 4,096, 16,384, and 65,536. All 36 Streaming cells return exactly the same ordered top- K key list and score array as their Full references; ordered keys and scores are also identical across chunk sizes. There are zero set-mismatch queries, zero tie-permutation queries, and zero R@10 mismatches. The tie-fixed Full rerun has macro R@10 of 42.1017%, and every Streaming cell reproduces its corresponding task-level R@10 exactly.

Streaming audit summary. Exactness is evaluated against the matched Full reference for each task; generation overhead is aggregate Streaming time relative to Full.

Chunk	Ordered keys	Score arrays	R@10 mismatches	Peak candidate buffer	Gen. overhead
4,096	12/12	12/12	0	$\leq 4,096$	+11.52%
16,384	12/12	12/12	0	$\leq 16,384$	+10.86%
65,536	12/12	12/12	0	$\leq 65,536$	+11.05%

The largest exposed frontier in this campaign contains 78,585 complete keys, so the 4,096-entry configuration exercises chunking on genuinely larger frontiers rather than merely matching the natural frontier size. The largest observed frontier P95 is 9,135.8 keys. The audit therefore establishes identical ordered key lists, score arrays, and R@10 values while bounding the complete-key candidate buffer by the chosen chunk size. The Python-level Streaming implementation is about 11% slower than Full, so we use it as a deterministic bounded-memory realization; the main timing results use the Full path.

Table S14: Default 5.61M-item UNION same-run control ($B = K = 50$, $s = 3$). Generation is measured in milliseconds per query from the precomputed-query-embedding boundary defined above.

Task	Seq. R@10	P2L R@10	Δ	Seq. ms/q	P2L ms/q	Reduction
CIRR-7	38.25	41.44	+3.19	50.08	25.70	48.7%
EDIS-2	40.54	50.23	+9.69	50.22	27.09	46.1%

Table S15: Default batch-1 results over five repeats. Latency statistics exclude the first 100 requests of each run; R@10 uses the complete query set. Means are ms/query $\pm$ sample standard deviation over per-run means; P95 is shown as Sequential/P2L.

Task	R@10 S/P	Gen. Seq.	Gen. P2L	Gen. P95 S/P	D2R Seq.	D2R P2L	D2R P95 S/P
CIRR-7	42.13/45.40	176.78 $\pm$ 3.80	102.01 $\pm$ 2.16	203.91/126.78	187.62 $\pm$ 3.89	112.64 $\pm$ 2.12	215.13/138.05
NIGHTS-4	48.54/50.14	176.60 $\pm$ 1.39	101.98 $\pm$ 1.76	203.54/127.83	186.91 $\pm$ 2.03	112.89 $\pm$ 1.45	214.47/138.88
EDIS-2	41.10/50.73	182.81 $\pm$ 1.38	106.43 $\pm$ 1.08	209.11/132.08	193.33 $\pm$ 1.92	117.21 $\pm$ 1.37	220.45/144.05
WebQA-1	32.26/46.64	183.97 $\pm$ 3.65	107.53 $\pm$ 1.77	211.62/133.23	194.87 $\pm$ 3.24	117.75 $\pm$ 1.58	223.35/144.18

Across the four tasks, generation falls by 41.6–42.3%, D2R by 39.4–40.0%, and D2R P95 by 34.7–35.8%. The semantic-search difference is 74.9–76.6 ms/query, numerically accounting for essentially the entire generation difference; the other measured stages remain similar. This localizes the measured saving without causally separating deferred selection from suffix conditioning.

Peak allocated device memory is identical between policies within each task: 2,607.120 MiB on CIRR-7, 2,595.108 MiB on NIGHTS-4, 2,601.677 MiB on EDIS-2, and 2,597.071 MiB on WebQA-1. Across repeats, the maximum reserved memory is 3,716, 3,704, 3,712, and 3,708 MiB, respectively; the largest within-task policy difference is 2 MiB. These are PyTorch device statistics, not process RSS measurements.

Table S16: Default P2L complete-leaf frontier from the batch-1 full-query logs. Distribution values are identical across five repeats; enumeration time is CPU seconds for the complete query set, reported as mean $\pm$ run SD. Cache denotes the continuation-cache footprint.

Task	Mean	P50	P95	P99	Max	Enumeration (s)	Cache (MiB)
CIRR-7	305.0	292	505.6	624.6	791	1.671 $\pm$ 0.041	1.01
NIGHTS-4	490.5	479.5	845.1	1,013.9	1,382	0.997 $\pm$ 0.022	1.87
EDIS-2	10,542.4	8,638	25,718	36,899	60,344	4.485 $\pm$ 0.021	46.47
WebQA-1	9,204.6	8,521	18,039.2	21,753.4	27,047	2.708 $\pm$ 0.028	23.39

EDIS-2 and WebQA-1 expose much wider descendant sets than CIRR-7 and NIGHTS-4. Nevertheless, Table S15 shows lower D2R mean and P95 at the measured default point on all four tasks. These measurements characterize this operating regime rather than a universal frontier-size threshold.

S4.2 Stage and UNION Frontier Accounting

Table S17: Batch-1 stage breakdown in ms/query over the same five repeats as Table S15. Each cell gives Sequential/P2L means. Generation includes framework overhead beyond the instrumented GPU stages; bucket lookup and reranking use CPU wall-clock timers.

Boundary	Stage	CIRR-7	NIGHTS-4	EDIS-2	WebQA-1
Within generation	Query input	23.79/23.76	23.69/23.84	20.68/20.68	20.93/21.04
	RQ pipeline	6.72/6.71	6.58/6.65	6.50/6.53	6.59/6.62
	Projector	0.21/0.21	0.21/0.21	0.21/0.21	0.21/0.21
	Semantic search	134.28/59.39	134.25/59.29	137.33/60.94	137.98/61.34
	Finalize	0.18/0.18	0.18/0.18	0.18/0.19	0.18/0.19
After generation	Bucket lookup	0.49/0.48	0.50/0.49	0.52/0.54	0.53/0.52
	Rerank	10.23/10.04	9.70/10.30	9.88/10.13	10.26/9.60

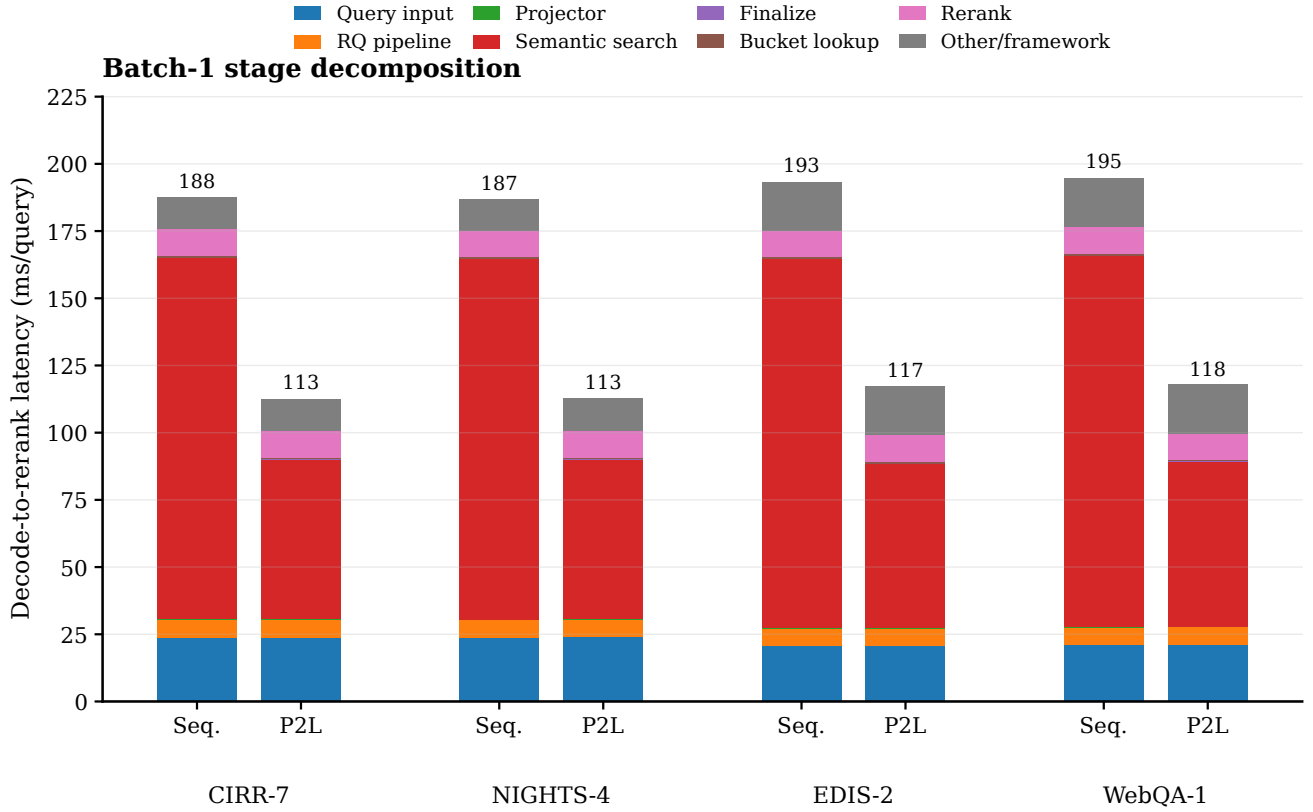


Figure S3: Batch-1 D2R stage decomposition over the same five-repeat campaign. The residual “Other/framework” segment closes each stack to the measured D2R mean because the instrumented GPU stages do not capture all framework overhead. The dominant policy difference is the semantic-search stage.

Table S18: Quality, generation latency, and exposed-frontier statistics for the fixed-policy CIRRR scale study. Times are batch-amortized ms/query; counts are complete keys. The 5.61M row is the separately measured matched UNION endpoint.

Items	R@10 (%)	Gen. mean	Gen. P95	Keys mean	Keys P95	Keys P99	Keys max
21,551	45.35	23.68	25.61	305.0	506.0	626.9	791
0.10M	45.06	24.27	25.97	301.6	503.0	620.9	799
0.50M	44.46	24.35	26.07	308.0	506.0	636.3	871
1.00M	44.08	24.14	25.83	320.1	521.0	655.3	1,563
5.61M	41.44	25.70	27.51	432.5	890.7	1403.9	6,513

Table S18 combines four fixed-policy nested-scale runs with a separately recorded 5.61M matched UNION endpoint. Each pool retains all CIRRR candidates and adds a fixed-seed prefix of UNION distractors. The main paper’s Figure 6(b–c) plots latency and frontier means/P95; this table adds quality and frontier tails for the tested collections.

Table S19: Persistent resources for the CIRRR scale collections and the separately measured 5.61M-item UNION endpoint. Candidates, distinct keys, and Trie nodes are in millions except the first row. Payload and vectors are MiB. Payload includes the nested Trie and cached code tensors; vectors are stored separately for the fixed reranker.

Candidates	Keys	Nodes	Payload	Vectors
21,551	21,214	125,796	3.67	31.57
0.10M	0.10M	0.61M	17.16	146.48
0.50M	0.50M	2.87M	84.71	732.42
1.00M	1.00M	5.54M	168.32	1,464.84
5.61M	5.50M	28.13M	927.32	8,216.42

These frontier distributions come from the same Independent-Suffix, default UNION runs used in the main comparison, not historical alternative-scoring runs. The CPU enumeration timer covers cached continuation lookup and parent-row creation. The GPU gather timer covers suffix-score gathering, summation, and complete-key row assembly. The top- K timer also includes gathering selected keys. Concatenation and host-to-device transfer remain inside generation but outside these GPU component timers, so their sum is not a complete continuation time.

UNION frontier counts describe distinct keys, not expanded items. Their P95/P99 are workload statistics, not output

Table S20: Frontier and instrumented continuation components from the same default UNION runs as the main quality–time comparison. Times are per-query averages over one full run, not batch-1 request latencies.

Task	Mean keys	P95	P99	Max	Enum. (ms)	Gather (ms)	Top- K (ms)
CIRR-7	432.5	890.7	1403.9	6,513	0.597	0.392	0.063
EDIS-2	11,096.8	27,881.0	38,530.2	59,294	1.765	0.462	0.106

budgets or guarantees. At the maximum EDIS-2 frontier in the UNION run, storing 59,294 nine-code keys as 64-bit integers and one FP32 score per key accounts for 4.30 MiB. This is analytical storage for two arrays; parent indices, intermediate gathers, cached continuations, overlapping tensor lifetimes, and batched queries add further temporary storage.

S4.3 Scorer-Independent Index Resources

The serialized payload contains the nested Candidate Trie, tokenized identifiers, candidate codes, and a cache signature. Prefix and node counts are computed from stored identifiers; file sizes are measured from the evaluated cache. These quantities do not depend on suffix-score configuration.

Table S19 reports persistent index resources. Candidate codes use nine 64-bit integers per item: their uncompressed accounting is $72N/2^{20}$ MiB for N candidates, already included in the serialized payload. The measured serialized payload sizes do not represent the resident size of the Python Trie.

For the 5.61M index, an isolated loader process takes 19.98 seconds to deserialize the nested Trie and cached code tensors, reaching 8,744.47 MiB peak RSS. This startup microbenchmark excludes the neural model, query features, and reranker embeddings, and Trie loading is excluded from decoder timing. The 8,216.42 MiB reranker feature tensor is stored separately; its simultaneous residency with the loader peak was not measured.

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